



computing
conference 2026

FairSwarm

A Cooperative Multi-Agent RL Approach for Fair Dynamic Pricing

Vishwanath Vinod, Rachel Kalpana Kalaimani

Indian Institute of Technology Madras, Chennai, India

.



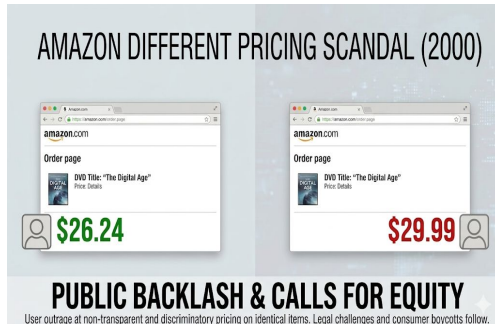
Fairness: The *Achilles' Heel* of Dynamic Pricing

Dynamic pricing **maximizes revenue** by adapting prices to demand and customer willingness to pay, potentially causing **price discrimination**.



Fairness: The *Achilles' Heel* of Dynamic Pricing

Dynamic pricing **maximizes revenue** by adapting prices to demand and customer willingness to pay, potentially causing **price discrimination**.



Coldplay reject dynamic pricing for 2025 UK tour in the wake of Oasis ticketing row



Fairness: The *Achilles' Heel* of Dynamic Pricing

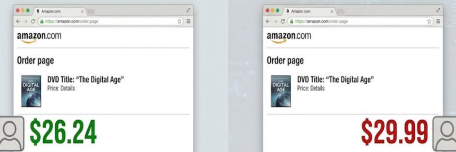
Dynamic pricing **maximizes revenue** by adapting prices to demand and customer willingness to pay, potentially causing **price discrimination**.



GLOBAL REGULATION IMMINENT FOR PREDATORY PRICING ALGORITHMS

Market fairness and equity prioritized as dynamic pricing faces 'unprecedented' public and regulatory backlash.

AMAZON DIFFERENT PRICING SCANDAL (2000)



PUBLIC BACKLASH & CALLS FOR EQUITY

User outrage at non-transparent and discriminatory pricing on identical items. Legal challenges and consumer boycotts follow.



Coldplay reject dynamic pricing for 2025 UK tour in the wake of Oasis ticketing row



How We Propose to Tackle It?

We address fairness in dynamic pricing with diverse customer distributions under finite inventory and limited selling horizon constraints².

1. **Algorithm:** Novel adaptation of CoMARL for Fair Dynamic Pricing.
2. **Dual Reward Structure:** Separate reward signals for Fairness and Profit.
3. **Tunable Control Parameter (λ):** Regulates the Fairness-Profit tradeoff.
4. **Empirical Validation:** Benchmarking against **baseline**³ methods in simulated multi-group markets.

² A. V. den Boer and B. Zwart, “Dynamic pricing and learning with finite inventories,” Operations research, vol. 63, no. 4, pp. 965–978, 2015

³ R. Maestre, J. Duque, A. Rubio, and J. Arevalo, “Reinforcement learning for fair dynamic pricing,” in Intelligent Systems and Applications: Proceedings of the 2018 Intelligent Systems Conference (IntelliSys) Volume 1. Springer, 2019, pp. 120–135.



Fairness Model: Metric not Constraint

- Fairness is modelled as an objective and not a constraint.
- Metric: **Jain Index**⁴ (*Resource allocation theory, Dynamic Pricing*)
- Range: [0,1] (higher value $\Rightarrow$ greater fairness)

⁴ R. Jain, A. Duresi, and G. Babic, "Throughput fairness index: An explanation," in ATM Forum contribution, vol. 99, no. 45, 1999, pp. 1-13.



Fairness Model: Metric not Constraint

- Fairness is modelled as an objective and not a constraint.
- Metric: **Jain Index**⁴ (*Resource allocation theory, Dynamic Pricing*)
- Range: [0,1] (higher value $\Rightarrow$ greater fairness)

$$f_{jain}(p_1, p_2, \dots, p_{|G|}) = \frac{\left(\sum_{g \in G} \bar{p}_g\right)^2}{|G| \sum_{g \in G} \bar{p}_g^2}$$

p_g : price offered to customer group g , where $g \in G$ and $|G|$ is the number of groups.

⁴ R. Jain, A. Duresi, and G. Babic, "Throughput fairness index: An explanation," in ATM Forum contribution, vol. 99, no. 45, 1999, pp. 1-13.



Limitation of Existing Methods³

- **Single Mixed Objective:** Fairness and profit combined into a weighted reward.

$$r_g = \begin{cases} \beta_p \cdot \tilde{p}_g + \beta_f \cdot f_{jain}(p_1, \dots, p_{|G|}) & \text{if Sale} \\ -0.5 & \text{if Penalty} \end{cases}$$

β_f and β_p : tunable parameters $\tilde{p}_g$: price normalized to $[0,1]$

³ R. Maestre, J. Duque, A. Rubio, and J. Arevalo, “Reinforcement learning for fair dynamic pricing,” in Intelligent Systems and Applications: Proceedings of the 2018 Intelligent Systems Conference (IntelliSys) Volume 1. Springer, 2019, pp. 120–135.



Limitation of Existing Methods³

- **Single Mixed Objective:** Fairness and profit combined into a weighted reward.

$$r_g = \begin{cases} \beta_p \cdot \tilde{p}_g + \beta_f \cdot f_{jain}(p_1, \dots, p_{|G|}) & \text{if Sale} \\ -0.5 & \text{if Penalty} \end{cases}$$

β_f and β_p : tunable parameters $\tilde{p}_g$: price normalized to $[0,1]$

- **Entangled Objectives:** Unfactorable during training.
- Reward Entanglement $\rightarrow$ Policy Oscillation $\rightarrow$ Unstable learning.
- Actions effect on Fairness vs Profit is not clearly distinguished.

³ R. Maestre, J. Duque, A. Rubio, and J. Arevalo, “Reinforcement learning for fair dynamic pricing,” in Intelligent Systems and Applications: Proceedings of the 2018 Intelligent Systems Conference (IntelliSys) Volume 1. Springer, 2019, pp. 120–135.



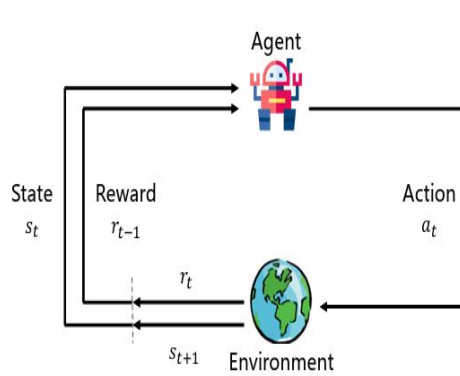
Formulating the Problem Using Multi-Agent RL

- **Markets** → Customer Groups: Each with distinct demand functions.
- **RL Agent**: One dedicated agent per market to maximize profit.

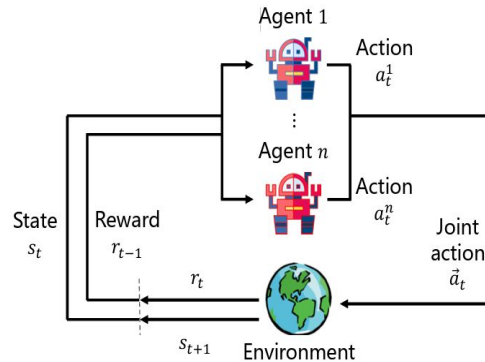


Formulating the Problem Using Multi-Agent RL

- **Markets** → Customer Groups: Each with distinct demand functions.
- **RL Agent**: One dedicated agent per market to maximize profit.



RL



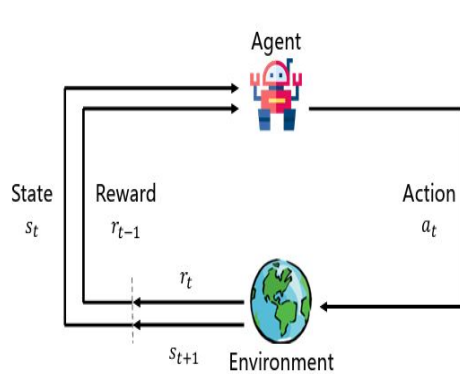
MARL

$$\text{Action: } a = \{\tilde{p}_1, \dots, \tilde{p}_{|G|}\}$$

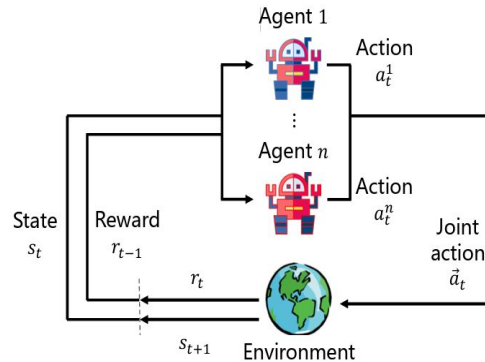
$$\text{State: } s = \{(d_1, n_1), \dots, (d_{|G|}, n_{|G|})\}$$

Formulating the Problem Using Multi-Agent RL

- **Markets** → Customer Groups: Each with distinct demand functions.
- **RL Agent**: One dedicated agent per market to maximize profit.



RL



MARL

$$\text{Action: } a = \{\tilde{p}_1, \dots, \tilde{p}_{|G|}\}$$

$$\text{State: } s = \{(d_1, n_1), \dots, (d_{|G|}, n_{|G|})\}$$

- **Policy (π_g)**: Pricing Strategy Learnt by RL Agent (unique for each market).
- **Episode Termination**: Inventory Exhausted or Selling Horizon ends.

Algorithm: Dual-Reward & Dual-Critic Architecture⁵

Individual Agent's Reward: Profit

$$r_g = \begin{cases} \tilde{p}_g, & \text{if Sale} \\ -0.5, & \text{if Penalty} \end{cases}$$

Depends only on its own Actions

Team reward: Fairness

$$\hat{r} = f_{\text{jain}}(p_1, \dots, p_{|G|})$$

Depends on Actions of all Agents

⁵ **H. U. Sheikh** and **L. B"ol"oni**, "Multi-agent reinforcement learning for problems with combined individual and team reward," in 2020 International Joint Conference on Neural Networks (IJCNN). IEEE, 2020, pp. 1–8.



Algorithm: Dual-Reward & Dual-Critic Architecture⁵

Individual Agent's Reward: Profit

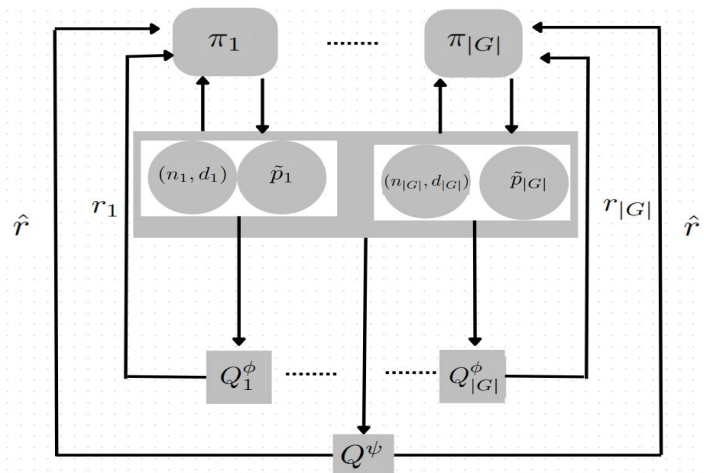
$$r_g = \begin{cases} \tilde{p}_g, & \text{if Sale} \\ -0.5, & \text{if Penalty} \end{cases}$$

Depends only on its own Actions

Team reward: Fairness

$$\hat{r} = f_{\text{jain}}(p_1, \dots, p_{|G|})$$

Depends on Actions of all Agents



- Separates the learning process.
- Local Critics (Q_g^ϕ)
 - Specific to each agent.
 - Estimates **Profit** (Individual reward).
- Global Centralized Critic (Q^ψ)
 - Shared by all Agents.
 - Input: Entire action and state space.
 - Estimates **Fairness** (Team reward)

⁵ H. U. Sheikh and L. B"ol"oni, "Multi-agent reinforcement learning for problems with combined individual and team reward," in 2020 International Joint Conference on Neural Networks (IJCNN). IEEE, 2020, pp. 1-8.



Algorithm: Policy Update

- Update policy (pricing strategy) using **Policy Gradient Step**.
- Combines Reward Signals from Global (fairness) and local critics (profit).



Algorithm: Policy Update

- Update policy (pricing strategy) using **Policy Gradient Step**.
- Combines Reward Signals from Global (fairness) and local critics (profit).

Conventional MARL Policy Gradient

$$\nabla_{\theta_i} J(\theta_i) = \mathbb{E}_{s,a \sim \mathcal{B}} [\nabla_{\theta_i} \pi_i(a_i | o_i) \nabla_{a_i} Q_i(s, a)]$$

FairSwarm Dual-Reward MARL Policy Gradient

$$\begin{aligned} \nabla J(\theta_g) = & \lambda \cdot \overbrace{\mathbb{E}_{s,a \sim \mathcal{B}} [\nabla_{\theta_g} \pi_g(\tilde{p}_g | (d_g, n_g)) \nabla_{\tilde{p}_g} Q^\psi(s, a)]}^{\text{Global Critic}} \\ & + \underbrace{\mathbb{E}_{(d_g, n_g), \tilde{p}_g \sim \mathcal{B}} [\nabla_{\theta_g} \pi_g(\tilde{p}_g | (d_g, n_g)) \nabla_{\tilde{p}_g} Q_g^{\phi_g}((d_g, n_g), \tilde{p}_g)]}_{\text{Local Critic}} \end{aligned}$$



Simulating Markets & Customer Interaction

Multiple Markets (Groups): Each with distinct demand function, capturing diverse customer behaviours.

$D_g(p_g) \sim \text{Bernoulli}(\mathbb{D}_g(p_g)) \forall p_g \in [p_l, p_h], g \in G$
 $(\mathbb{D}_g(p_g))$: Value of Demand Function at price p_g

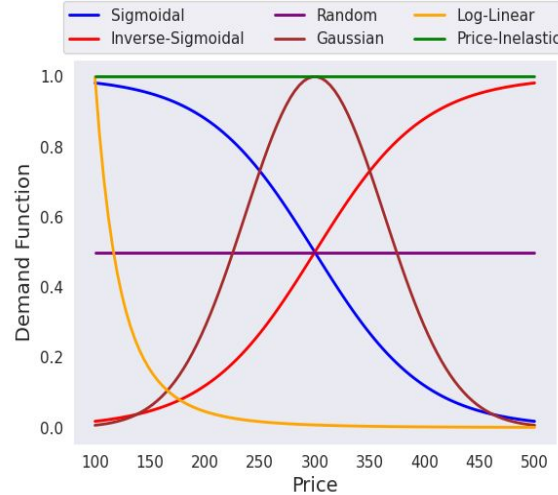


Table 1: Demand Functions for Customer Groups

Group	Customer Behaviour	Demand Function ($\mathbb{D}_g(p)$)
Group 1	Sigmoidal Demand	$\left(1 + e^{-(b+w \cdot p)}\right)^{-1}$
Group 2	Inverse Sigmoidal Demand	$\left(1 + e^{(b+w \cdot p)}\right)^{-1}$
Group 3	Log-Linear Demand	$e^{-(b' + w' \cdot \ln(p))}$
Group 4	Price Inelastic Demand	1 (constant)
Group 5	Gaussian Demand	$e^{-\left(\frac{(p-\mu)^2}{2\sigma^2}\right)}$
Group 6	Random Demand	0.5 (constant)

Parameters: $b = 6$ $w = -0.02$ $\mu = 300$
 $b' = -20.44$ $w' = 4.44$ $\sigma^2 = 8000$

Simulating Markets & Customer Interaction

Multiple Markets (Groups): Each with distinct demand function, capturing diverse customer behaviours.

$$D_g(p_g) \sim \text{Bernoulli}(\mathbb{D}_g(p_g)) \quad \forall p_g \in [p_l, p_h], g \in G$$

$(\mathbb{D}_g(p_g))$: Value of Demand Function at price p_g

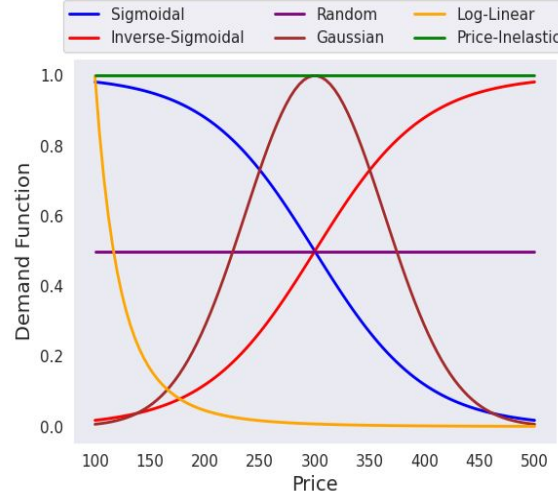
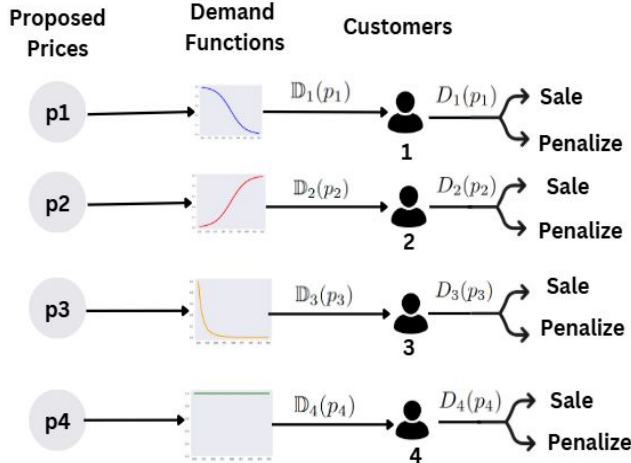


Table 1: Demand Functions for Customer Groups

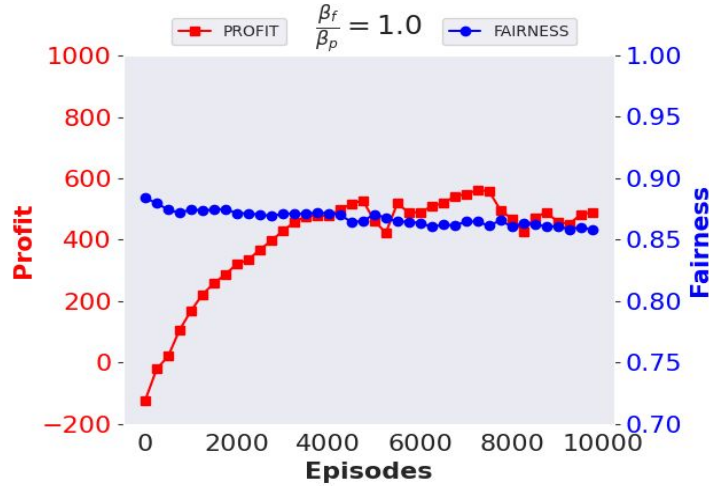
Group	Customer Behaviour	Demand Function ($\mathbb{D}_g(p)$)
Group 1	Sigmoidal Demand	$\left(1 + e^{-(b+w \cdot p)}\right)^{-1}$
Group 2	Inverse Sigmoidal Demand	$\left(1 + e^{(b+w \cdot p)}\right)^{-1}$
Group 3	Log-Linear Demand	$e^{-(b' + w' \cdot \ln(p))}$
Group 4	Price Inelastic Demand	1 (constant)
Group 5	Gaussian Demand	$e^{-\left(\frac{(p-\mu)^2}{2\sigma^2}\right)}$
Group 6	Random Demand	0.5 (constant)

Parameters: $b = 6$ $w = -0.02$ $\mu = 300$
 $b' = -20.44$ $w' = 4.44$ $\sigma^2 = 8000$

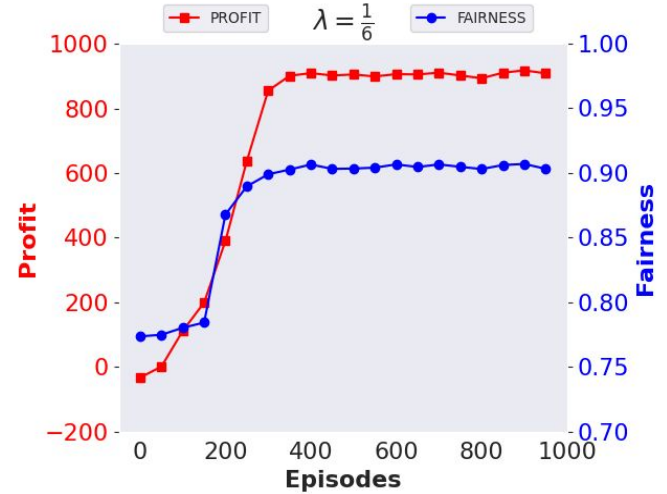
- Per Timestep of episode: One customer interaction per market.
- **Assumption:** Constant Customer Influx (to isolate demand disparities).

$$\text{Interaction: } \begin{cases} \text{Sale} & \text{if } D_g(p_g) = 1 \\ \text{Penalty} & \text{if } D_g(p_g) = 0 \end{cases}$$

FairSwarm converges to **higher levels of Profit & Fairness** with **Faster Convergence**.



(a) 6 Group: Baseline

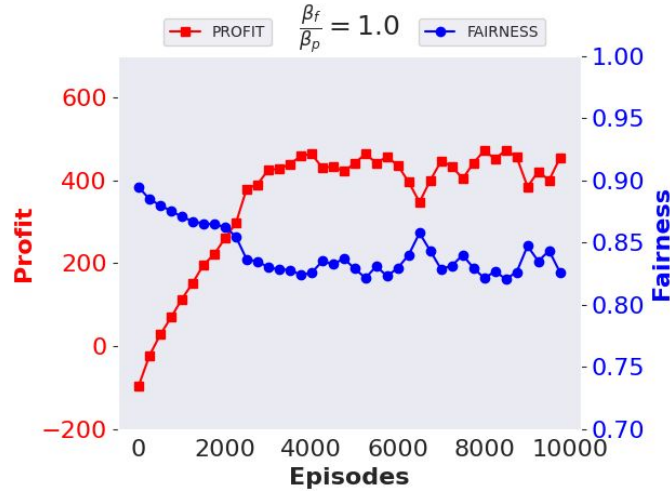


(b) 6 Group: FairSwarm

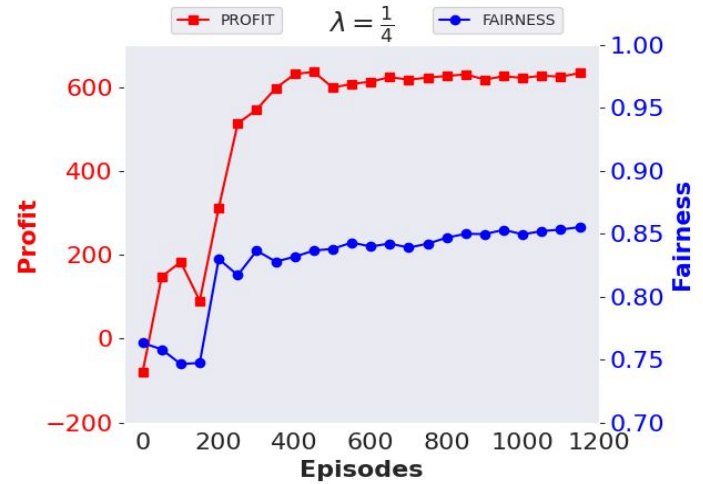
Comparison of **profit** and **fairness** against baseline for selected customer distributions consisting 2, 4 and 6 groups from Table 1 with $\frac{\beta_f}{\beta_p} = 1.0$ and $\lambda = \frac{1}{|G|}$



FairSwarm converges to **higher levels of Profit & Fairness** with **Faster Convergence**.



(a) 4 Group: Baseline

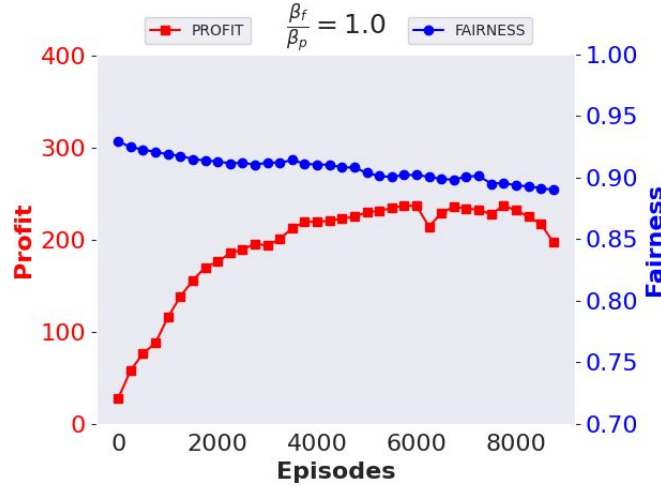


(b) 4 Group: FairSwarm

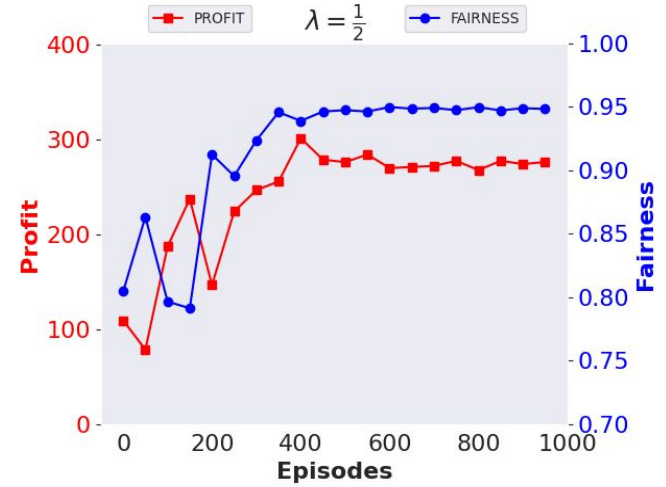
Comparison of **profit** and **fairness** against baseline for selected customer distributions consisting 2, 4 and 6 groups from Table 1 with $\frac{\beta_f}{\beta_p} = 1.0$ and $\lambda = \frac{1}{|G|}$



FairSwarm converges to **higher levels of Profit & Fairness** with **Faster Convergence**.



(a) 2 Group: Baseline



(b) 2 Group: FairSwarm

Comparison of **profit** and **fairness** against baseline for selected customer distributions consisting 2, 4 and 6 groups from Table 1 with $\frac{\beta_f}{\beta_p} = 1.0$ and $\lambda = \frac{1}{|G|}$



Control Parameter (λ) Effectively Controls **Fairness**-**Profit** Tradeoff

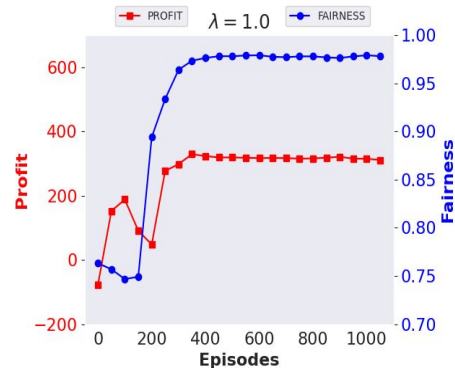
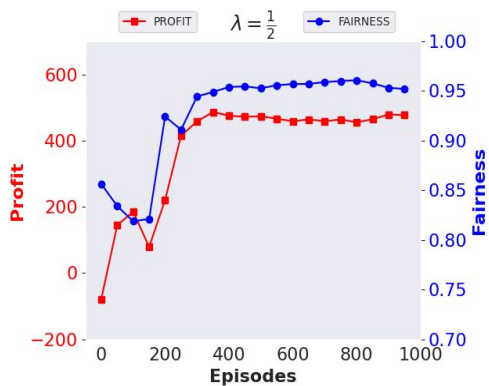
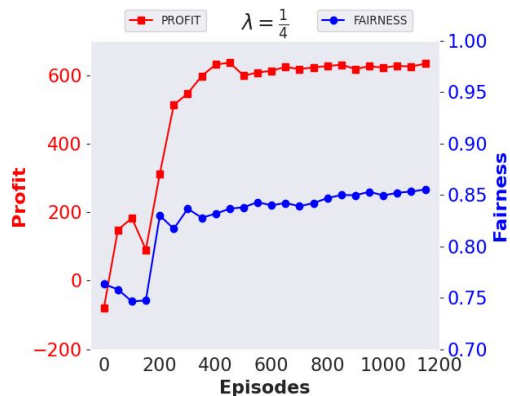
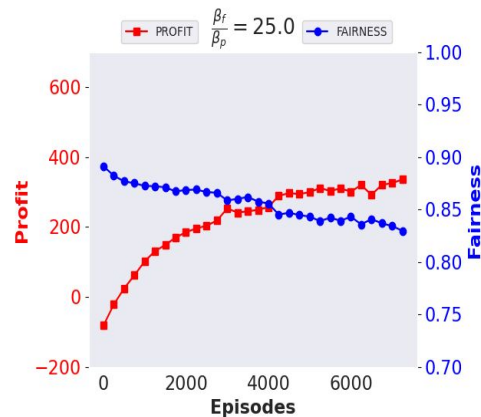
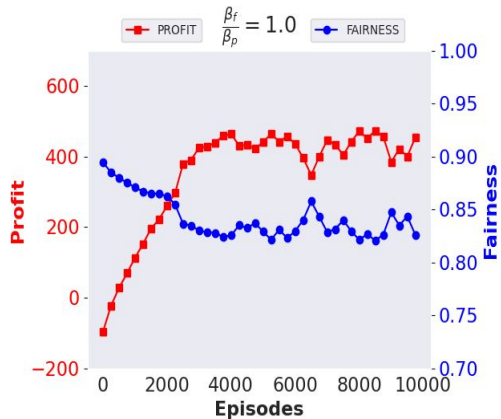
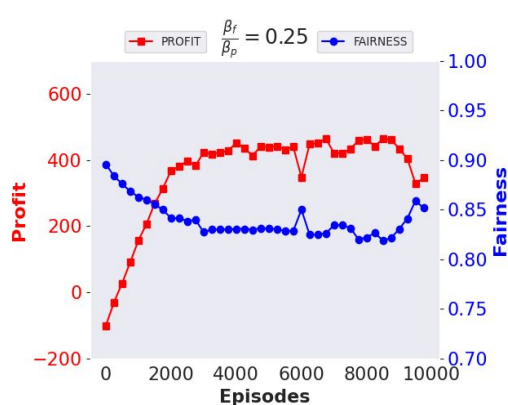
Higher λ prioritizes fairness over profit and vice versa.

Table 2: Profit and Fairness after 1000 Episodes Averaged over Three Trials for Different Group and λ Settings

Groups (G)	λ	Profit	Fairness
1,2	$\frac{1}{ G }$	259.6 ± 20.1	0.952 ± 0.004
	1	105.7 ± 30.3	0.983 ± 0.006
1,2,3,4	$\frac{1}{ G }$	590.6 ± 49.9	0.819 ± 0.007
	0.5	391.1 ± 65.1	0.958 ± 0.005
	1	313.9 ± 8.3	0.981 ± 0.004
1,2,3,4,5,6	$\frac{1}{ G }$	917.2 ± 33.0	0.921 ± 0.008
	0.5	826.9 ± 30.1	0.934 ± 0.004
	1	622.2 ± 4.1	0.964 ± 0.001

λ is a better control parameter than $\frac{\beta_f}{\beta_p}$ for **fairness**-**profit** tradeoff.

$$\frac{\beta_f}{\beta_p}$$



$$\lambda$$

Conclusion

- The proposed dual-reward CoMARL framework results in:
 - Joint Optimization to Higher Levels of fairness and profit.
 - Stable Learning & Faster Convergence.
 - Effective tuning of fairness-profit trade off using λ .

Future Work

- Relax constant customer influx assumption.
- Explore policy distillation⁶ or distributed optimization techniques.
- Evaluation on real-world market data.

⁶ L. Wang, Y. Zhang, Y. Hu, W. Wang, C. Zhang, Y. Gao, J. Hao, T. Lv, and C. Fan, “Individual reward assisted multi-agent reinforcement learning,” in Proceedings of the 39th International Conference on Machine Learning.



Contacts:



Vishwanath Vinod,
Undergraduate (B.Tech),
[Department of Electrical Engineering @ IIT Madras.](#)
Mail: vishwanathvinod2004@gmail.com



Rachel Kalpana Kalaimani,
Associate Professor, Control and Optimization Group,
[Department of Electrical Engineering @ IIT Madras.](#)
Mail: rachel@ee.iitm.ac.in

